Comparative Intestinal Permeability and Molecular Responses to Major Aquafeed Mycotoxins in RTgutGC Cells

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Introduction

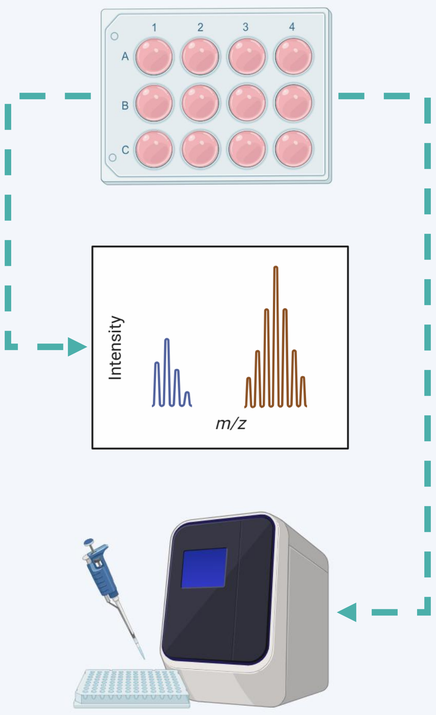
Incorporation of cereals into aquafeeds increases the risk of **mycotoxin contamination** [1]. Mycotoxins are fungal metabolites that affect **fish health** and can **bioaccumulate** in edible tissues, posing **a food safety concern** [1].

The mechanisms of **mycotoxin intestinal absorption** and their effects on the fish intestinal barrier remain poorly understood [2]. **RTgutGC** cell line, the first established intestinal fish cell line, is used as an ***in vitro* model of the fish intestinal barrier** [3]. **Cell models** provide a greener, more ethical and reproducible alternative to *in vivo* trials [3].

Using the **RTgutGC** cell line model, this study evaluated barrier **permeability** and **molecular responses** to aflatoxin B1 (AFB1), fumonisin B1 (FB1), enniatin B (ENNB) and enniatin B1 (ENNB1) to:

- Assess the suitability of the model for mycotoxin studies;
- Investigate their toxicological impacts in fish;
- Provide insights for aquaculture risk assessment and food safety.

Methodology



Intestinal transport assay RTgutGC

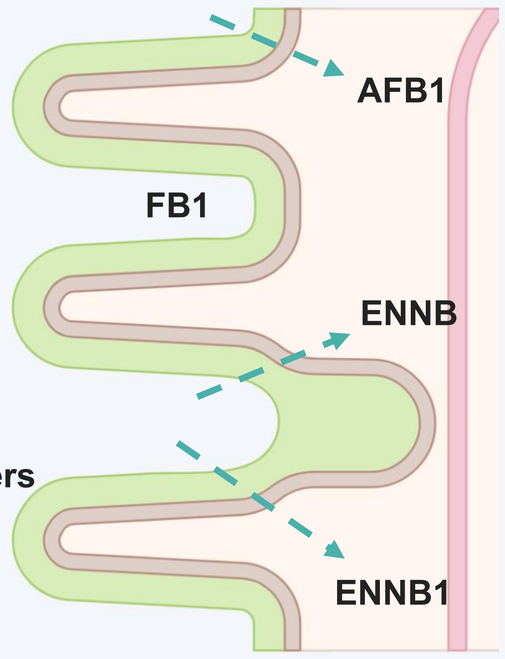
62500 cells/cm²
28-day differentiation
4 independent assays
400 µg/L - Isolated and in combination
3, 12 and 24 h

Mycotoxin quantification

QuEChERS microextraction
LC-MS/MS

qPCR Genetic expression of **key markers**

Epithelial barrier function
Immune response
Stress response



Results and discussion

Transport assay

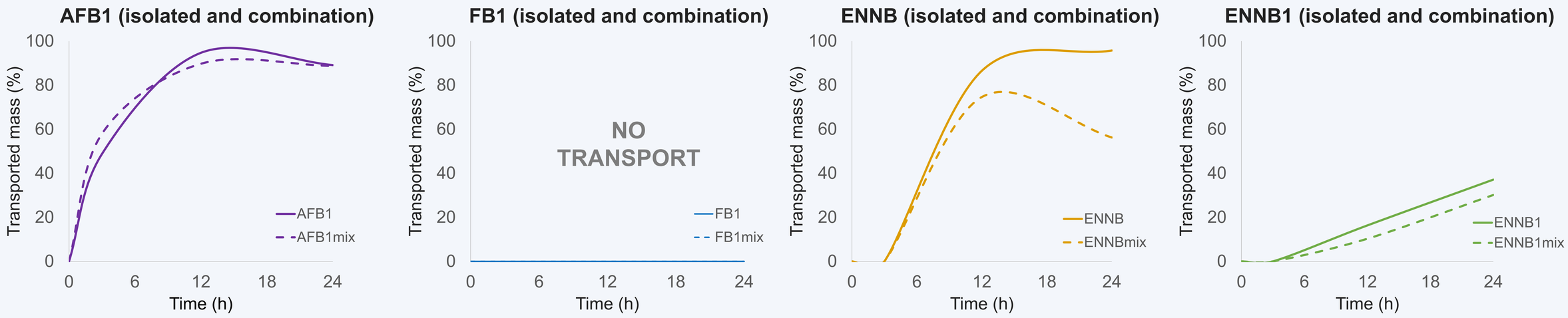


Figure 1 – Transported mass (%) of mycotoxins aflatoxin B1 (AFB1), fumonisin B1 (FB1), enniatin B (ENNB) and enniatin B1 (ENNB1) over time (24 h) across the epithelial barrier in the intestinal transport assay (n= 4) using an *in vitro* cell model with RTgutGC cells (62500 cells/cm²) grown in permeable Transwell inserts (1.1 cm²) with 28-day differentiation in Leibowiz-15 medium supplemented with 5% heat-inactivated foetal bovine serum (FBS), 1% penicillin/streptomycin and 1% glutamine.

Gene expression

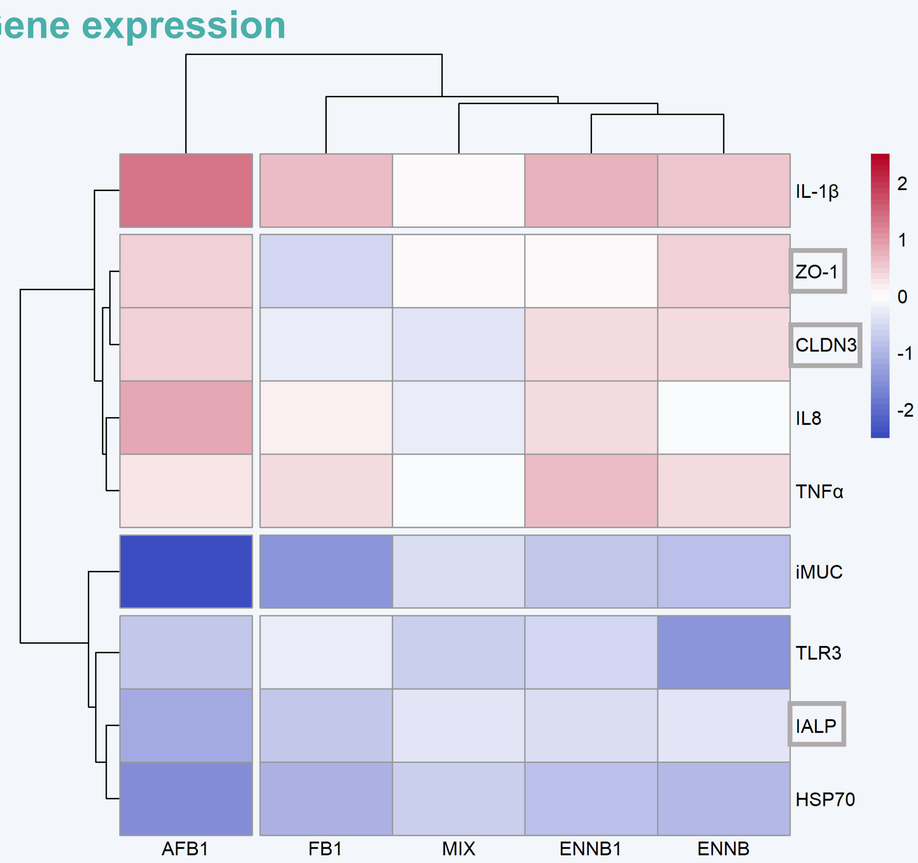


Figure 2 – Hierarchical cluster with dendrogram and heatmap of mycotoxin exposure treatments (AFB1, FB1, Mix, ENNB1 and ENNB) and barrier (marked with grey blocks), immune and stress response genes (IL-1β, interleukin 1β; ZO-1, zonula occludens 1; CLDN3, claudin 3; IL8, interleukin 8, TNFα, tumour necrosis factor α; iMUC, mucin; TLR3, toll like receptor 3; IALP, intestinal alkaline phosphatase, and HSP70, heat shock protein 70). Clustering analysis was performed by measuring the Euclidean distance between data. Heatmap represents normalized gene expression values (log²), where the genes are in the rows and the mycotoxin treatments in the columns. The colour range displays up-regulated (red) and down-regulated (blue) values, representing deviations in gene regulation.

Transport across the intestinal barrier

AFB1, ENNB and ENNB1 were efficiently **transported** at different rates. FB1 was **not transported**.

AFB1 and ENNB1 showed **similar patterns with isolated or combined exposure**. AFB1 crosses **rapidly initially**, followed by a slower increase after 12 h of exposure. At 24 h, there is a **slight decrease**, potentially indicating **efflux**.

ENNB, when **combined** with the other mycotoxins, showed a **decrease in transported mass** at 24h, indicating **potential efflux transport**.

ENNB1 shows a steady transport rate; however, it has the **lowest transported mass** at 24 h.

Molecular responses to mycotoxin exposure

All mycotoxins induced similar overall regulation of barrier, immune and stress markers.

AFB1 exposure resulted in more discernible stress responses:

- ↑ **IL-1β**, a pro-inflammatory cytokine and mediator of innate immune response
- ↓ **iMUC**, a mucus marker relevant in fish immune response

FB1 exposure resulted in a slightly different response:

- ↓ **of both ZO-1 and CLDN3**, tight-junction genes essential for maintenance of barrier function

Exposure to the **combined mycotoxins** resulted in a slight **down-regulation** response. Clearly demonstrating how the **presence of one contaminant may affect the action of another**.

Conclusions

- AFB1, ENNB and ENNB1 were transported across the **RTgutGC** intestinal barrier while FB1 was not.
- Exposure to all mycotoxins induced molecular responses, **disrupting the intestinal barrier, and activating immune response and stress related genes**. AFB1 was the most active mycotoxin.
- The ***in vitro* cell model** proved its efficacy as a predictive tool for assessing **mycotoxin bioavailability and toxicity** in fish. The results obtained provide new insights for **risk assessment** in aquaculture and **food safety**.

References: [1] Lorusso et al., 2025, doi: 10.3390/toxins17070356; [2] Koletsi et al., 2021, doi: 10.3390/toxins13060403; [3] Minghetti et al., 2017, doi: 10.1007/s10565-017-9385-x.

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